

REASONING ARTIFACT

The Dimensional Invariant:

Sleep Duration, Sleep Topology, Attractor Basin Depth, and Long-Term Survival Across the Entire History of Life

The oldest surviving lineages sleep the least.

The minimum of the sleep spectrum is the maximum of the survival spectrum.

The topology of sleep payment is a geometric prediction of the landscape.

The cost of navigation is ultimately an oxidative cost.

This is not a coincidence. This is the geometry.

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DERIVATION NOTICE AND VERSION HISTORY

v1.0 (doi:10.5281/zenodo.19094935): Established the dimensional invariant: sleep duration tracks basin-occupation cost; basin depth predicts survival horizon; the oldest surviving lineages sleep the least. Three cost axes (Axis 1 between-basin switching, Axis 2 within-basin metabolic, Axis 3 within-basin computational intensity). Two survival strategies (Lock, Navigator). The unstable zone. Six predictions (P-INVARIANT-1 through P-INVARIANT-6).

v2.0 (doi:10.5281/zenodo.19101495): Made explicit that sleep topology — the architecture of the discharge process — is itself a geometric prediction of the landscape, not a measurement artefact. Five topologies derived (T1 through T5). Full three-channel sonar (duration, topology, behavioural flexibility). Added P-INVARIANT-7 (single-celled organisms). Added framing note: the topology list is open; new topologies are confirmations of geometric generativity, not falsifications.

v3.0 (this document): Three new geometric findings derived from exploratory research against the current literature, each producing content that changes the formal statement of the invariant:

(1) **The oxidative cost substrate.** Sleep deprivation kills via reactive oxygen species (ROS) accumulation in peripheral high-metabolism tissue (Vaccaro et al., *Nature*, 2020; confirmed through 2024). Basin-occupation cost is not only synaptic — it is fundamentally oxidative. This gives the framework a biochemical floor and explains the ectothermic Navigator topology (T2) geometrically: lower metabolic rate produces ROS at a rate that distributed rest can clear, making concentrated NREM unnecessary.

(2) **Two new topologies.** T6: Transition-cost sleep. Sleep is required not to discharge ongoing occupation cost but to pay the cost of a developmental basin transition (*C. elegans* lethargus; broadly, animal metamorphosis). T7: Immune-basin sleep. Pathogen invasion recruits the organism into an immune-response basin with its own occupation cost, discharged as sickness quiescence. Both topologies are geometrically derivable from first principles and confirmed empirically.

(3) **Population-level basin portfolio.** Bacterial persister cells demonstrate the dimensional invariant operating at population level: a single species simultaneously maintains a Lock sub-population (persister cells, deep metabolic quiescence, perturbation-invariant) and a Navigator sub-population (active majority, moderate metabolic cost, flexible response). The population holds a portfolio of basin depths rather than a single basin position. This extends the invariant below the organism level entirely.

FRAMING NOTE: The seven topologies derived here are not a complete taxonomy of sleep architectures. They are the topologies derivable from the constraint set known at the time of this derivation. The

framework predicts that any new topology discovered in nature will be derivable from the same geometric principles: the interaction of navigational cost structure, oxidative cost rate, metabolic architecture, environmental lethality constraints on full sleep, developmental transition requirements, and immune-basin occupation. A new topology is not a falsification of the framework. It is a confirmation of the geometric generativity of the framework. The framework is not a list. It is a geometry. Geometries are not made obsolete by new solutions. They are confirmed by them.

THE DIMENSIONAL INVARIANT

Across all life, at all scales, across all of deep time, the attractor geometry framework ($S + N + G \rightarrow R$) makes three inseparable predictions about rest:

Prediction 1 — Quantity: The total cost of basin occupation must be discharged per cycle. Cost is decomposable into three axes: Axis 1 (between-basin switching cost), Axis 2 (within-basin metabolic cost), Axis 3 (within-basin computational intensity). Duration of discharge tracks total cost. Basin depth predicts survival horizon. The oldest surviving lineages carry the lowest cost. They discharge the least.

Prediction 2 — Topology: The architecture of the discharge process is itself a geometric prediction of the landscape position, not an independent biological variable. Seven topologies are derived. Each is the only architecturally feasible solution for its corresponding landscape position given the full constraint set.

Prediction 3 — Substrate: The cost of basin occupation is fundamentally oxidative. Navigation generates reactive oxygen species as a metabolic byproduct. Sleep discharges both the synaptic cost (via slow-wave downscaling and neural replay) and the oxidative cost (via ROS clearance in peripheral high-metabolism tissues) simultaneously. The topology of sleep determines the efficiency and architecture of both discharge processes together.

The oldest surviving lineages sleep the least. How they sleep is the geometry of where they are. What they are paying is an oxidative debt.

1. The Framework

1.1. The Attractor Geometry Framework

$$S + N + G \rightarrow R$$

- S = structural substrate (genome and encoded build capacity)
- N = coherence gradient field (external or internal field specifying basin occupation)
- G = global attractor landscape (basin depths, barrier heights, accessible states, topology of transitions)
- R = resulting phenotype or state

The framework predicts that an organism's position in G simultaneously determines:

1. The total cost of occupying that position — what must be discharged.
2. The survival trajectory — how resistant the basin is to perturbation.
3. The topology of the discharge process — what architecture of rest is geometrically required.
4. The biochemical substrate of the cost — oxidative, synaptic, or both, in proportions determined by the metabolic architecture of the organism.

These four are projections of a single geometric object: the organism's full position in the landscape G .

1.2. The Core Claim

The metabolic cost of occupying a position in an attractor landscape is a function of the navigational demand imposed by that position. Navigational demand is determined by: (1) the number of accessible basins; (2) barrier heights between basins; (3) frequency of required transitions; (4) the depth of Basin Ω if one exists; and (5) the within-basin metabolic or computational intensity of the occupied basin.

Rest is the generic process by which accumulated cost is discharged. **The cost is fundamentally oxidative.**

Navigation consumes metabolic substrate and generates reactive oxygen species as a byproduct in proportion to the rate and intensity of neural and somatic activity. Sleep deprivation kills via ROS accumulation in peripheral high-metabolism tissues — specifically the intestinal epithelium in *Drosophila* (Vaccaro et al., *Nature*, 2020), which is the tissue most directly responsible for substrate acquisition driving the navigation.

The synaptic and oxidative discharge processes are not independent. They are co-extensive: slow-wave sleep performs synaptic downscaling via the Tononi–Cirelli mechanism and simultaneously drives glymphatic pulsation (CSF flow through perivascular AQP4 channels) that clears neurotoxic metabolites. Both processes require the same sleep state. The sleep topology must be sufficient to discharge both costs simultaneously or in rapid alternation.

The architecture of the rest process — its topology — is not a separate biological variable. It is a further projection of the same landscape geometry, constrained by the metabolic architecture of the organism and the environmental conditions that determine whether any given topology is energetically and ecologically feasible.

1.3. Why the Existing Theoretical Frameworks Are All Correct and Incomplete

The four dominant sleep function theories in the current literature are:

1. Synaptic Homeostasis Hypothesis (Tononi & Cirelli): sleep downscale synaptic potentiation. Correct. Describes the synaptic component of Axis 1 and Axis 3 cost discharge.
2. Energy conservation: sleep reduces metabolic demand. Correct. Describes the consequence of reduced navigational activity during Axis 2 cost recovery.
3. Glymphatic clearance (Nedergaard): sleep drives CSF pulsation clearing neurotoxic waste. Correct. Describes the physical mechanism by which the oxidative cost substrate is cleared during Topology 3 discharge.
4. Memory consolidation: sleep stabilises memory traces. Correct. Describes a downstream consequence of synaptic downscaling — the net effect of selectively strengthening versus weakening synapses during discharge.

None of these frameworks generates the dimensional invariant. None predicts that the oldest surviving lineages sleep the least. None predicts convergent sleep topologies across independent phylogenies. None predicts that sleep topology is a geometric property of landscape position. None connects sleep duration to deep-time survival trajectories.

They are correct descriptions of mechanisms within Topology 3 (consolidated bilateral deep sleep in endothermic organisms). The dimensional invariant is the geometric structure that contains all four as special cases.

2. The Cost Structure

2.1. The Three Axes

1. **Axis 1 — Between-basin switching cost.** The organism navigates between multiple accessible basins. Each transition requires bilateral synaptic reconfiguration. Cost accumulates continuously during waking. Produces both synaptic cost (hemispheres must reconfigure for new basin) and oxidative cost (high neural firing rates during contextual switching generate ROS). *Geometric signature:* High behavioural flexibility. Multiple accessible basins. High sleep. *Examples:* Felidae, Basin Ω organisms.
2. **Axis 2 — Within-basin metabolic cost.** A single expensive basin consumes all available metabolic resources during waking. Sleep is required for metabolic resource recovery. Primarily an oxidative cost rather than synaptic: the organism's peripheral tissues accumulate ROS and metabolic debt during sustained high-cost waking activity in a single specialised basin. *Geometric signature:* Low behavioural flexibility. Single basin. High sleep from resource depletion. *Example:* Koala (eucalyptus detoxification is a continuous oxidative cost).
3. **Axis 3 — Within-basin computational intensity.** Extreme neural computation within a single specialised basin. High firing rates generate high ROS in neural tissue. Synaptic reconfiguration cost from processing intensity, not switching. Both synaptic and oxidative costs are high. *Geometric signature:* Low behavioural flexibility. Extreme within-basin performance. High sleep. *Example:* Brown bat (echolocation at millisecond resolution for 4 hours generates neural ROS and synaptic cost requiring 20 hours of discharge).

The Lock organism operates on none of these axes. Zero navigational demand. Zero within-basin metabolic surplus cost. Zero within-basin computational intensity. Minimal ROS generation from minimal activity. The sleep minimum is the signature of zero cost at every axis and zero oxidative debt simultaneously.

2.2. The Oxidative Cost Substrate: Why This Matters Geometrically

The identification of ROS accumulation as the proximate death mechanism in sleep deprivation (Vaccaro et al., 2020) is not merely a mechanistic finding. It is a geometric finding, because it tells you what the cost of navigation actually is at the biochemical level.

The geometric consequence: The feasibility of any sleep topology is constrained by the rate at which the organism generates ROS during its active period and the rate at which the chosen topology can clear that ROS.

An endothermic Navigator generates ROS at endothermic metabolic rates during 8–16 hours of active navigation. The ROS accumulation rate is high. Distributed rest (Topology 2) cannot clear it fast enough because glymphatic pulsation requires sustained NREM pressure to drive adequate CSF flow. Concentrated bilateral deep sleep (Topology 3) is required.

An ectothermic Navigator generates ROS at ectothermic metabolic rates — an order of magnitude lower per unit time. The ROS accumulation rate is low enough that extended distributed bilateral rest (Topology 2) clears it adequately without requiring concentrated NREM. This is the geometric explanation for why the crocodilian uses Topology 2 rather than Topology 3. It is not a taxonomic accident. It is the only feasible topology for an ectotherm with Navigator-level ecological complexity.

The prediction that follows: Any ectothermic Navigator will use Topology 2 or a topology with lower NREM concentration than endothermic Navigators. Any organism shifted from ectothermy to endothermy (e.g., via experimental manipulation of body temperature regulation) will show a shift toward Topology 3 and increased sleep concentration. This is a novel, testable, geometrically derived prediction.

3. The Seven Sleep Topologies

The framework predicts that different landscape positions produce different sleep architectures. These topologies are geometric predictions, not empirical categories. Each is the only architecturally feasible discharge solution for its corresponding landscape position given the full constraint set (navigational cost structure, ROS generation rate, metabolic architecture, environmental lethality of full sleep, developmental requirements, immune-basin structure).

T1: Metabolic quiescence / biochemical rhythm only

Landscape position: Deepest Lock. No nervous system or minimal neural architecture. Zero navigational demand. Zero synaptic reconfiguration cost. Minimal ROS generation from minimal activity.

Discharge: No sleep in any neurological sense. Circadian biochemical rhythms regulate metabolic processes. ROS cleared by antioxidant enzyme cycling on circadian schedule without requiring behavioural quiescence.

Duration: Undefined or approaching zero in neurological terms.

Geometric signature: Maximum perturbation-invariance. If an organism shows T1, it is at or near the extreme Lock extreme of the survival spectrum.

Confirmed examples: Sponge (Porifera, ~600 Myr). Horseshoe crab (Chelicerata, ~450 Myr). Most jellyfish (Cnidaria, 500–700 Myr).

T2: Distributed bilateral rest — ectothermic Navigator solution

Landscape position: Moderate Navigator with ectothermic metabolic architecture. Navigator-level navigational cost (multiple accessible basins, moderate switching cost) but ROS generation rate is an order of magnitude lower than endothermy. Energy budget for full bilateral wakefulness is limited per day.

Discharge: True bilateral EEG deep sleep for 2–6 hours discharging synaptic cost. Extended distributed bilateral rest state (reduced neural activity, bilateral EEG, behavioural vigilance via brainstem alerting, not cortical asymmetry) for 12–16 hours clearing the low-rate oxidative debt continuously. Total Navigator-equivalent discharge distributed across the inactive period.

Why bilateral and not unihemispheric: Navigator switching cost is between ecological basins, not between

waking hemispheres. Both hemispheres carry equivalent cost and discharge simultaneously. Unihemispheric asymmetry is not required because the oxidative and synaptic costs are bilaterally symmetric.

Why distributed and not concentrated: Low ectothermic ROS generation rate means glymphatic clearance does not require sustained high-pressure NREM to achieve adequate flow. Distributed low-pressure rest clears the oxidative debt adequately. Concentrated NREM is metabolically expensive and ecologically dangerous for an ectotherm in an open environment.

Duration: 2–6h true sleep plus 12–16h Basin C rest. Total equivalent: Navigator range.

Confirmed examples: Crocodilian (Archosauria, ~200 Myr). Kelly & Lesku (2015, 2019): bilateral EEG confirmed throughout rest periods. One-eye-open vigilance is behavioural (brainstem-level alerting), not unihemispheric in EEG sense.

Predicted additional examples: Monitor lizards and large ectothermic generalist reptiles. Large ectothermic generalist fish. Cockroach (Blattodea, ~300 Myr) — ectothermic Navigator, 12h distributed rest.

T3: Consolidated bilateral deep sleep — endothermic solution

Landscape position: Endothermic organism at any cost level. High ROS generation rate during waking (endothermic metabolic rates) requires concentrated high-efficiency glymphatic clearance during sustained NREM. Bilateral synaptic reconfiguration cost discharged during same concentrated window.

Discharge: Concentrated NREM slow-wave sleep driving high-pressure glymphatic pulsation (CSF flow through AQP4 perivascular channels) to clear neurotoxic metabolites and amyloid-beta. Simultaneous synaptic downscaling via slow waves. REM phase discharges computational replay cost (Axis 3) and consolidates navigational memory.

Duration: 2–22 hours depending on cost axis. The full range from minimal Lock (2–5h) to extreme specialist (20h) is expressed within this topology. Duration within T3 is a valid direct proxy for total cost because the metabolic architecture is held constant (endothermy).

Confirmed examples: All endothermic organisms: rodent (~13h), human (7–9h), koala (~20h), brown bat (~20h), lion (~16h), elephant (~2h), giraffe (~30min when not suppressed).

Note on cross-topology comparison: Duration is a valid proxy for cost comparison *within* T3 only. A crocodilian (T2, 2–6h true sleep) and an elephant (T3, 2h) have similar true sleep durations but completely different cost structures. The topology must be identified before duration is interpreted.

T4: Permanent Basin Ω — dissolved waking/sleep barrier

Landscape position: High navigational cost in an environment where full bilateral sleep is lethal (open ocean mandatory ventilation, multi-directional predation in continuous motion). The cost of entering the sleep basin's exit barrier exceeds the accumulated synaptic and oxidative cost of remaining partially awake indefinitely.

Discharge: The barrier between waking and sleep is dissolved. One hemisphere sleeps (synaptic downscaling, partial glymphatic pulsation) while the other remains active (navigation, motor control, sensory processing), in continuous alternation. Each hemisphere discharges in rotation. Oxidative cost clearance is continuous and distributed at the hemispheric level.

Why unihemispheric: The navigational cost and oxidative cost are bilaterally symmetric (same as T2), but the lethal constraint on full bilateral sleep forces the discharge to be asymmetric. T4 is T2 with an additional lethal constraint that forces the distributed rest to be hemispheric rather than whole-brain.

Duration: ~8h total equivalent but never bilateral. Continuous hemispheric rotation.

Confirmed examples: Dolphin, porpoise, some seals. Migratory birds during peak flight (partial T4 via unihemispheric microsleeps). Frigatebirds during multi-day oceanic flight.

T5: Fragmented shallow sleep — external suppression with rebound

Landscape position: Deep single basin (low navigational cost, low within-basin cost) but chronic external pressure (predation, ecological constraint) continually interrupts sleep before discharge can complete. The sleep basin is behaviourally shallow, not geometrically shallow. The organism has navigational cost and oxidative debt that accumulates but cannot be discharged fully in any single bout.

Discharge: Brief fragmented bilateral sleep bouts. Incomplete discharge per bout. Debt accumulates across sessions and is discharged as sleep rebound when external pressure is removed.

Diagnostic signature: Sleep rebound under safe conditions is the definitive identifier of T5. An organism in T1 or a true Lock in T3 will show no rebound when pressure is removed. An organism in T5 will show immediate and disproportionate sleep increase, paying accumulated debt.

Duration: 30 min to 2h total per day under pressure. Rebound may produce 12–16h concentrated sleep immediately post-pressure.

Confirmed examples: Giraffe. White-crowned sparrow during migration shows a *partial* T5-adjacent state (sleep reduction without rebound), but this is better classified as basin restructuring: the migratory state is a different attractor with lower navigational cost than the non-migratory state. See Section 7.

T6: Transition-cost sleep — developmental basin crossing

*NEW IN v3.0. Derived from *C. elegans* lethargus data and generalised from first principles.*

Landscape position: An organism at a developmental basin transition point. The organism is not discharging ongoing occupation cost. It is paying the cost of a *transition* between two distinct landscape basins (larval stages, metamorphic transitions, critical developmental windows).

Geometric structure: In the attractor landscape G , the transition between two basins has a cost that is distinct from the occupation cost of either basin. The transition requires wholesale reconfiguration of neural circuitry, synaptic architecture, and metabolic programming. This cost cannot be discharged during normal waking activity because waking activity *is* the reconfiguration event. The organism must be in a quiescent state to allow the reconfiguration to complete without being interrupted by ongoing navigational demand.

Discharge: Mandatory quiescence (lethargus in *C. elegans*). Duration proportional to the magnitude of the basin transition, not to the organism's navigational cost profile. A large transition (complete metamorphosis in holometabolous insects) requires longer quiescence than a small developmental stage transition.

Diagnostic signature: Sleep occurs at fixed developmental timepoints regardless of prior waking activity. No rebound after transition completes — the debt was transition cost, not waking occupation cost. Sleep cannot be postponed without developmental failure; this is the definitive distinction from T3 and T5.

Confirmed examples: *C. elegans* lethargus at each larval stage transition (Allada & Siegel, 2008; confirmed through 2024). Larval-to-pupal transition sleep in *Drosophila*. Predicted broadly in all holometabolous insects and animals with discrete developmental basin transitions.

Novel prediction: The depth of quiescence in T6 scales with the magnitude of the basin transition, not with the organism's ongoing navigational demand. A highly specialised organism undergoing a major metamorphic transition will show longer T6 quiescence than a generalist organism undergoing the same transition, because the specialised basin it is transitioning into is deeper and the transition cost is proportionally higher.

T7: Immune-basin sleep — defensive basin occupation

*NEW IN v3.0. Derived from *C. elegans* sickness quiescence data and generalised from first principles.*

Landscape position: The organism's landscape has been invaded by a perturbation (pathogen, toxin, severe injury) that creates a new high-cost basin in G : the immune response basin. The immune response basin is not the organism's standard ecological basin. It is a temporary defensive attractor that consumes metabolic resources at very high rates (fever, cytokine production, immune cell proliferation) and generates very high ROS.

Geometric structure: Entering the immune response basin has a basin-occupation cost that must be discharged to allow recovery. The organism cannot simultaneously navigate its ecological basins and pay the immune-basin cost. Sleep is the process by which the immune-basin cost is discharged: reduced navigational demand frees metabolic resources for immune function, and sleep-specific molecular processes (cytokine-driven slow-wave enhancement, immune cell trafficking during NREM) are the discharge mechanism.

The field's framing: Sickness sleep is a “side effect” of cytokine signalling on sleep-promoting circuits. *The geometric framing:* Sickness sleep is the organism paying the occupation cost of the immune response basin. Cytokines are the basin-entry signal — they are N in the $S + N + G \rightarrow R$ equation, the coherence gradient field specifying which basin the organism must occupy.

Diagnostic signature: Sleep duration scales with immune response intensity, not with prior navigational demand. Duration returns to baseline as pathogen is cleared and immune basin occupation cost falls to

zero. T7 sleep cannot be prevented without impairing immune function — this is the definitive distinction from all other topologies.

Confirmed examples: *C. elegans* sickness quiescence induced by *Pseudomonas aeruginosa* (2024 biorxiv; multiple prior confirmations). Human and mammalian fever sleep. Insect sickness quiescence. Confirmed across all major animal phyla where tested.

Novel prediction: The magnitude of T7 sleep scales linearly with the depth of the immune response basin (measurable by cytokine titre, inflammatory marker concentration, or metabolic rate increase during illness). Organisms with deeper ecological basins (Lock organisms) will show proportionally larger T7 responses relative to their baseline sleep because their baseline is so low that any immune basin cost represents a large fractional increase. Lock organisms sleeping near-zero under normal conditions will show dramatic sleep induction under pathogen challenge. This is testable in horseshoe crabs, which show circadian immune activation cycles but have near-zero baseline sleep.

The seven topologies span the full geometric range of discharge architectures predicted by the framework. They are not a closed taxonomy. They are the currently derivable solutions given the known constraint set. The framework predicts their existence from first principles. Any new topology found empirically is a confirmation, not a challenge.

4. The Two Survival Strategies

4.1. Strategy 1: The Lock

Basin structure	Single basin. Extremely deep. Very high exit barrier.
Morphology	Extreme conservatism. Body plan unchanged across hundreds of millions of years.
Metabolism	Slow. Efficient. Minimal resource requirement. Minimal ROS generation.
Behaviour	Low flexibility. Narrow repertoire. Highly programmatic.
Sleep quantity	Minimal. Approaching zero in simplest forms. 2–5 hours in complex Lock organisms.
Sleep topology	T1 (quiescence/biochemical rhythm) in simplest forms. T2 or T3 at low duration in neurally complex Lock organisms. T7 (immune-basin) when infected — predicted to show dramatic T7 response relative to near-zero baseline. Never T4 (no open-ocean lethal constraint). Never T6 unless undergoing developmental transition.
Oxidative profile	Near-zero ROS generation from near-zero navigational activity. Antioxidant cycling sufficient without behavioural quiescence.
Survival horizon	Maximum. 300–600 Myr+.
Survival mechanism	Basin so deep it is perturbation-invariant. Extinction events cannot displace the organism.
Convergence	Seven independent phyla.

4.2. Strategy 2: The Navigator

Basin structure	Multiple basins. Shallow barriers. High switching capacity.
Morphology	Ecological generalism. Flexible body plan.
Metabolism	Flexible. Moderate. Endothermic or ectothermic. Metabolic architecture determines which topology discharges the Navigator cost.
Behaviour	High flexibility. Broad repertoire. Contextual switching.
Sleep quantity	Moderate equivalent. Total Navigator-range discharge cost is consistent across Navigators but delivered through different topologies depending on metabolic architecture.
Sleep topology	Ectothermic Navigator: T2 (distributed bilateral rest, low ROS rate, no NREM concentration required). Endothermic Navigator: T3 (consolidated bilateral, high ROS rate requiring concentrated glymphatic clearance). Open-ocean Navigator with lethal bilateral constraint: T4 (permanent Basin Ω). All Navigators: T7 when infected. T6 if undergoing developmental transition.
Oxidative profile	Moderate to high ROS generation from moderate to high navigational activity. Topology selected to match clearance rate to generation rate.
Survival horizon	Long. 100–300 Myr.
Survival mechanism	Navigates to alternative basin when current basin disappears. Generalism is itself Basin Ω of ecological flexibility.
Convergence	Six+ independent lineages across T2, T3, and T4.

4.3. The Unstable Zone

Basin structure	Shallow single basin. High barriers to alternative basins.
Morphology	Narrow specialist. Not deep enough to be perturbation-invariant. Not flexible enough to reach alternatives.
Sleep quantity	Variable. Often high (Axis 2 or Axis 3) with no survival advantage.
Sleep topology	Typically T3. Duration does not discriminate Unstable Zone from Navigator within T3. Discriminator is behavioural flexibility and dietary breadth, not sleep architecture.
Survival horizon	Short. Millions of years.
Examples	<i>T. rex</i> , mammoth, sabre-tooth cat, passenger pigeon, dodo.

4.4. Why The Lock Outlasts The Navigator

Basin depth is the ultimate survival parameter. Navigation is a survival strategy for moderate perturbations. For catastrophic perturbations that eliminate all accessible basins simultaneously, only the deepest basins survive. The Permian–Triassic event killed 96% of all marine species. The horseshoe crab (T1, near-zero sleep, near-zero navigational cost, near-zero ROS generation) was still there on the other side.

The minimum of the sleep spectrum is the maximum of the survival spectrum. The minimum of the oxidative cost is the minimum of the displacement cost of the extinction event. The geometry is the same at every scale.

5. The Population-Level Basin Portfolio

This section presents a finding derived from exploratory research in v3.0 that extends the dimensional invariant below the organism level.

5.1. Bacterial Persister Cells: The Dimensional Invariant at Population Level

Bacterial persister cells (reviewed in Niu et al., *Signal Transduction and Targeted Therapy*, 2024) are phenotypic variants within a clonal population that enter deep metabolic quiescence under stress. They are not genetically resistant to antibiotics. They are transiently in a state of near-zero metabolic activity that renders them immune to antibiotics that require active metabolic processes as targets. Upon stressor removal, persisters resuscitate and repopulate.

The geometric reading:

A clonal bacterial population under antibiotic challenge simultaneously maintains two distinct basin positions:

1. **Active majority (Navigator-equivalent):** Moderate metabolic activity. Flexible metabolic response. High ROS generation. Subject to antibiotic killing. Short individual survival horizon.
2. **Persister minority (Lock-equivalent):** Near-zero metabolic activity. Deep metabolic quiescence. Near-zero ROS generation. Perturbation-invariant to the antibiotic challenge. Maximum individual survival horizon for the duration of the challenge.

The population holds a **portfolio of basin depths** rather than a single basin position. This is the dimensional invariant operating as an evolutionary portfolio strategy at the sub-organismal population level.

The geometric prediction this generates:

The proportion of persisters in a population will scale with the historical frequency and intensity of perturbation events in the organism's ecological history. A population with a history of frequent antibiotic-equivalent challenges (high-perturbation environment) will maintain a higher persister fraction than a naive population. This is the population-level analogue of the Lock/Navigator trade-off at the individual level: populations in volatile environments evolve toward higher Lock fractions; populations in stable environments evolve toward lower Lock fractions and higher active fractions.

The further implication:

The same portfolio logic applies at the organism level in a different form. Basin Ω organisms (cats, dolphins, corvids) maintain the *capacity* for both basins simultaneously but express one at a time. The persister cell population maintains both basin positions simultaneously as distinct cellular phenotypes within the same clonal population. These are two implementations of the same geometric strategy: maintaining access to both deep (Lock-equivalent) and flexible (Navigator-equivalent) attractor positions simultaneously, with the allocation between them determined by the current perturbation environment.

6. Local Sleep: Circuit-Level Application of the Dimensional Invariant

Recent work (Nir & Krueger, 2024; Tononi & Cirelli, 2023) has demonstrated that sleep is not a global brain state but a collection of local events: individual cortical columns can cycle between sleep-like and wake-like states on a second timescale, independently of global sleep state. Brain regions more active during waking show higher slow-wave activity during subsequent sleep.

The geometric reading:

The dimensional invariant applies at circuit level. Individual cortical circuits each have their own basin-occupation cost that accumulates independently during waking. Circuits used more heavily during the day accumulate higher synaptic and oxidative cost. They discharge that cost locally during sleep, independently of other circuits.

The global slow wave is not a global reset. It is the statistically concurrent discharge of many local circuits that were used during the same waking period, appearing global because waking activity uses most circuits simultaneously. When waking activity is asymmetric (one hemisphere used more, one cortical region used more), the sleep response is correspondingly asymmetric.

This is a novel extension of the framework: The three cost axes apply at circuit level, not just organism level. A hand motor cortex region running Axis 3 (computational intensity) during a skilled motor task discharges that specific Axis 3 cost locally during subsequent sleep. The visual cortex discharges its own cost. The prefrontal cortex its own. All simultaneously but independently.

The prediction this generates:

The local sleep finding predicts that surgical removal or lesioning of a cortical region will reduce the corresponding component of whole-brain slow-wave activity during subsequent sleep, proportional to the navigational cost that region was carrying. A lesion to a high-Axis-3 region should reduce total NREM slow-wave power more than a lesion to a low-activity region. This is testable and has not been explicitly framed this way in the current literature.

7. Geometric Analysis of Apparent Divergences

7.1. The Migratory Sparrow: Basin Restructuring, Not Debt Suppression

White-crowned sparrows (*Zonotrichia leucophrys*) reduce sleep to ~30% of normal during migration and show no sleep rebound after migration terminates. P-INVARIANT-5 predicts that minimal sleep under pressure is

associated with T5 (rebound on pressure removal). The sparrow shows no rebound. Is this a falsification?

Geometric resolution:

The migratory state is not the same attractor as the non-migratory state. Migration recruits the organism into a distinct basin in G — the migratory attractor — with a different navigational cost structure than the residential ecological basin. In the migratory basin:

- Axis 1 switching cost is reduced: the organism is in a single focused basin (directional flight and fuelling) rather than navigating between multiple ecological basins.
- Axis 3 cost is redirected: neural computation is dominated by magnetoreceptive navigation rather than social and ecological decision-making.
- The sleep requirement in the migratory basin is genuinely lower than in the residential basin, not suppressed.

The no-rebound finding confirms this: rebound would only occur if debt had accumulated. No rebound means no debt. The sparrow was paying its actual (lower) migratory-basin cost through unihemispheric microsleeps during flight (T4-adjacent) and brief rest bouts at stopovers.

Refinement to P-INVARIANT-5: Sleep rebound identifies T5 (external suppression of unchanged navigational demand). It will *not* be observed when sleep reduction accompanies genuine basin restructuring that lowers actual navigational demand. The two cases are distinguishable by measuring whether performance within the current basin is maintained during the low-sleep period. T5: performance declines (debt accumulating). Basin restructuring: performance maintained within the new basin’s demands.

7.2. Ecological Generalism as Consequence: Bidirectional Reinforcement

A 2024 Nature study reported that ecological generalism may be more a consequence of widespread success than a direct cause of survival. This appears to challenge the Navigator framework.

Geometric resolution:

The dimensional invariant is not a cross-sectional claim about currently living species. It is a deep-time trajectory claim. The Nature 2024 finding describes a bidirectional reinforcement mechanism that the framework predicts and that makes the Navigator basin deeper over time:

Navigational success (survival of perturbations) → wider geographic range → more ecological contexts encountered → Navigator basin deepened through experience and population-level exploration → greater navigational success.

This is a positive feedback loop that deepens Basin Ω over evolutionary time. The cockroach has been running this loop for 300 Myr. Its current generalism is simultaneously the cause and consequence of 300 Myr of Navigator survival. The Nature 2024 finding is correct about the consequence direction. The dimensional invariant describes the full loop and its deep-time trajectory.

8. The Empirical Record

8.1. Lock Strategy: Confirmed Examples

Lineage	Age	Sleep quantity	Topology	Basin geometry
Sponge (Porifera)	~600 Myr	Undefined. No nervous system.	T1. Biochemical rhythm only.	Deepest possible. Single programme. Zero ROS from navigation.
Jellyfish (Cnidaria)	500–700 Myr	Undefined most. <i>Cassiopeia</i> ~8h.	T1 most. T2 border in <i>Cassiopeia</i> .	No brain. Radial symmetry. 500+ Myr unchanged.
Nautilus (Mollusca)	~500 Myr	Quiescent periods. No REM.	T1–T2 border. No hemispheric architecture.	Neutral buoyancy. Passive ambush. Shell unchanged.
Horseshoe crab (Chelicerata)	~450 Myr	Tidal/circadian rest only.	T1. No bilateral sleep.	Deepest terrestrial Lock. 450 Myr unchanged.
Shark (Chondrichthyes)	~400 Myr	Rest ~2–4h distributed.	T2 border. No REM confirmed.	Low navigational demand. Borders Lock/Navigator.

Coelacanth (Sarcoptery- gii)	~400 Myr	Unknown. Mini- mal presumed.	T1–T2. Cold wa- ter suppresses all activity.	Minimal metabolism. 400 Myr unchanged.
Tuatara (Lepi- dosauria)	~200 Myr	12h+ rest. Mostly torpor-like dor- mancy.	T2. Distributed ectotherm rest. Not true bilateral deep sleep.	Slowest reptile metabolism. 5°C function. 100+ year lifespan.

8.2. Navigator Strategy: Confirmed Examples

Lineage	Age	Sleep quantity	Topology	Basin geometry
Cockroach (Blattodea)	~300 Myr	~12h rest cycle.	T2. Distributed ectotherm. Bilat- eral EEG.	Eats anything. Lives anywhere. 300 Myr survival.
Crocodylian (Archosauria)	~200 Myr	2–6h true bilat- eral EEG sleep. 12–16h Basin C bi- lateral rest.	T2. Bilateral EEG confirmed (Kelly & Lesku, 2015, 2019). One-eye-open is behavioural, not unihemispheric EEG.	Ectothermic Navigator. Generalist predat- or. Survived K–Pg. Any prey, any water. Topology predicted from ROS rate and metabolic architecture.
Octopus (Cephalopoda)	~300 Myr	Quiet sleep + active (REM-like) sleep in cycles.	T3-equivalent. Two-phase dis- charge indepen- dently evolved. Convergent with vertebrate NREM/REM.	Navigator. Tool use. Problem solving. Dis- tributed neural archi- tecture. Independent convergence of T3.
Dolphin (Cetacea)	~50 Myr	~8h total. Never bilateral.	T4. Permanent Basin Ω . Uni- hemispheric alter- nation.	Open ocean. High cog- nition. Mandatory sur- face ventilation. Bilat- eral sleep lethal.
Rodent (Muri- dae)	~65 Myr	~13h.	T3. Consolidated bilateral. Stan- dard endothermic Navigator.	All biomes. All diets.
Corvidae	~30 Myr	~8–10h.	T3. Some T4 dur- ing roosting.	Tool use. Problem solv- ing. Social complexity.

8.3. Unstable Zone: Confirmed Examples

Lineage	Age	Quantity	Topology	Why extinct
<i>T. rex</i> (Tyranno- sauridae)	2–3 Myr	Unknown.	T3 presumed.	Narrow apex predator. No alternative basin at K– Pg.
Mammoth (Pro- boscidea)	0.5– 5 Myr	~2h (cf. ele- phant).	T3, T5 (preda- tion pressure).	Grassland specialist. Habitat contracted. Could not adapt diet.
Sabre-tooth cat (Machairo- dontinae)	0.5– 11 Myr	High felid- range.	T3. Basin Ω an- cestry.	Mega fauna specialist. Mega fauna gone. Basin collapsed.
Dodo (Raphi- dae)	0.003 Myr	~8–10h.	T3.	Island specialist. Single shallow basin. First per- turbation = extinction.

9. The Sleep Spectrum as Three-Channel Sonar

The complete sonar instrument has three channels. None of the three alone is sufficient to specify an organism’s landscape position. All three together triangulate it precisely.

Channel 1 — Duration: Measures total discharge cost. Valid proxy within a single topology class. Ambiguous across topology classes.

Channel 2 — Topology: Identifies which of the seven discharge architectures the organism uses. Removes topology-class ambiguity from duration. T1 and T2 produce similar apparent durations to true Lock; topology distinguishes them.

Channel 3 — Behavioural flexibility: Discriminates Navigator (high flexibility, multiple accessible basins) from Unstable Zone (low flexibility, single shallow basin) when both occupy T3. Discriminates T5 (flexibility impaired under pressure, debt accumulating) from basin restructuring (flexibility maintained within new basin).

Supplementary — Rebound response: Diagnostic for T5 vs. true Lock. Not needed once T1/T2 are identified by topology, but required to distinguish low-duration T3 from T5 when topology data is unavailable.

Supplementary — ROS rate: Predicted to discriminate ectothermic T2 from endothermic T3 at equivalent navigational cost levels. Testable via metabolic rate measurement during active period.

Duration	Topology	Horizon	Landscape position	Examples
~0	T1	600 Myr+	Deepest Lock. Zero navigational cost. Zero ROS from navigation.	Sponge, horse-shoe crab
0–4h	T1–T2	300–500 Myr	Deep Lock or minimal neural Lock. Low ROS.	Shark, coelacanth, nautilus
T2 distributed	T2	100–300 Myr	Ectothermic Navigator. Low ROS rate makes distributed rest sufficient.	Crocodilian, cockroach, tuatara
8h never bilateral	T4	50–200 Myr	Open-ocean Navigator. Lethal bilateral constraint. Basin Ω of waking/sleep.	Dolphin
8–14h consolidated	T3	10–100 Myr	Endothermic Navigator or social coupling. High ROS needs concentrated glymphatic clearance.	Rodent, corvid, human
0.5–2h fragmented	T5	Variable	External suppression. Rebound diagnostic. Debt accumulating.	Giraffe
14–22h	T3	Niche-dependent	Axis 2 metabolic, Axis 3 computational, or deep Basin Ω .	Koala, bat, lion
Fixed developmental	T6	N/A	Developmental basin transition. Duration $\propto$ transition magnitude, not waking cost.	<i>C. elegans</i> lethargus, insect metamorphosis
Scales with illness	T7	N/A	Immune basin occupation. Duration $\propto$ cytokine titre / immune response depth.	All animals under pathogen challenge

10. Convergence Across Independent Phyla

10.1. The Lock Converges Across Seven Independent Phyla

Phylum / Class	Example	Age	Topology
Porifera	Sponge	~600 Myr	T1
Cnidaria	Jellyfish	500–700 Myr	T1
Mollusca	Nautilus	~500 Myr	T1–T2
Chelicerata	Horseshoe crab	~450 Myr	T1
Chondrichthyes	Shark	~400 Myr	T2 border
Sarcopterygii	Coelacanth	~400 Myr	T1–T2
Lepidosauria	Tuatara	~200 Myr	T2

Seven independent arrivals at the same geometric strategy. The transition from T1 to T2 within the Lock as neural complexity increases is itself a geometric prediction: adding neural complexity adds synaptic cost to discharge and ROS from neural firing, but the basin remains deep enough to keep total cost minimal. The topology shifts but the duration remains in the Lock range.

10.2. The Navigator Converges Across Multiple Topologies in Six+ Independent Lineages

Lineage	Example	Age	Topology
Blattodea	Cockroach	~300 Myr	T2 (ectotherm)
Archosauria	Crocodylian	~200 Myr	T2 (ectotherm, bilateral EEG confirmed)
Cephalopoda	Octopus	~300 Myr	T3-equivalent (independently evolved two-phase discharge)
Cetacea	Dolphin	~50 Myr	T4 (lethal bilateral constraint)
Rodentia	Rat	~65 Myr	T3 (endotherm)
Aves	Corvid	~30 Myr	T3, partial T4 during roosting
Primates	Human	~6 Myr	T3 (endotherm, inter-generational Basin B*)

The Navigator strategy has independently converged across T2, T3, and T4 in at least seven lineages. The strategy is the same geometric object — ecological generalism as Basin Ω of ecological flexibility — expressed through the only topologically feasible discharge architecture available given each lineage’s metabolic and environmental constraints.

The octopus confirmation is among the strongest available: completely independent neural architecture, 500+ Myr of separate evolution, Navigator ecological strategy confirmed, and convergent two-phase sleep (quiet plus active REM-like) independently derived. The framework predicted this convergence. The topology is determined by the navigational demand, not by the biology.

10.3. The Unstable Zone Does Not Converge

Each extinction event eliminates different taxa via different mechanisms. No convergent feature exists in the unstable zone except instability itself. This non-convergence is a confirmation of the framework: there is no deep basin in the unstable zone to converge on.

11. The Human Position

The human is an endothermic Navigator in T3 (consolidated bilateral deep sleep, 7–9 hours). Navigator range. Standard endothermic discharge architecture. Nothing anomalous in the sleep signature.

The unprecedented feature is the inter-generational Basin B* — a co-constructed social attractor that deepens across generations through cultural accumulation. No other known organism has a basin whose depth is a function of time and intentional investment rather than purely of evolutionary selection.

The T7 (immune-basin) response in humans is well-documented and confirms the framework: fever sleep is mandatory and proportional to immune response intensity. Sleep deprivation during illness demonstrably impairs recovery — this is the immune-basin occupation cost not being discharged.

The T6 (transition-cost) response is observable in human development: neonatal sleep is dramatically elevated (16–18h) during the period of maximum neural basin formation. Sleep duration decreases monotonically as developmental transitions complete and the adult attractor landscape stabilises. This is the transition-cost topology declining as the basin transition work is completed, leaving only ongoing occupation cost to discharge.

The risk the framework identifies remains: a Navigator that is simultaneously the agent eliminating the alternative basins for all other species is undermining the conditions that make its own strategy viable. This is a geometric statement about landscape structure, not a moral one. The framework does not predict the trajectory of a species actively remodelling the landscape on which the Navigator strategy depends. This is the honest limit of the current derivation.

12. Literature Confirmations

12.1. Oldest Surviving Lineages Sleep the Least

Derived before literature. Confirmed: Horseshoe crab, nautilus, jellyfish, shark. All confirmed as minimal discharge, T1 or T2.

12.2. Crocodilian Sleep is Bilateral EEG, Not Unihemispheric

Derived: ectothermic Navigator predicts T2. *Confirmed:* Kelly & Lesku (2015, 2019). Bilateral EEG throughout rest periods. One-eye-open vigilance behavioural, not cortical hemispheric asymmetry.

12.3. Oxidative Cost is the Death Mechanism in Sleep Deprivation

Derived: basin occupation cost is fundamentally oxidative. *Confirmed:* Vaccaro et al. (*Nature*, 2020). ROS accumulation in gut epithelium is the proximate death cause. Antioxidant supplementation prevents death under total sleep deprivation. Confirmed and extended through 2024.

12.4. Developmental Transitions Require Mandatory Sleep

Derived: T6 transition-cost sleep predicted from basin transition cost principle. *Confirmed:* *C. elegans* lethargus at each larval transition. Allada & Siegel (2008); Cassada & Russell (1975); multiple confirmations through 2024. Sleep at developmental transitions is mandatory and cannot be postponed without developmental failure.

12.5. Octopus Independently Evolved Two-Phase Sleep

Derived: Navigator strategy predicts T3-equivalent two-phase discharge regardless of neural architecture. *Confirmed:* Medeiros et al. (2021, *iScience*). Octopus with quiet sleep plus active (REM-like) sleep. Confirmed and extended through 2023–2024. Complete independent convergence of two-phase discharge topology in a Navigator organism with entirely different neural architecture.

12.6. Mass Extinction Survivors Are Disproportionately Lock or Navigator

Confirmed: K–Pg survivors: crocodilians (Navigator, T2), turtles (partial Lock), birds (Navigator, T3/T4), small mammals (Navigator, T3). Permian–Triassic: horseshoe crab (Lock, T1), nautilus (Lock, T1). Source: Jablonski (1986); Barnosky et al. (2011).

12.7. Both Strategies Are Convergent Across Independent Phyla

Confirmed. Source: Conway Morris (2003, *Life's Solution*).

12.8. Generalist Diet Predicts Extinction Survival

Confirmed: Capellini et al. (2008, *Proc Royal Soc B*); Barnosky et al. (2011).

12.9. Single-Celled Organisms Show Metabolic Rest States Tracking Niche Breadth

Derived: P-INVARIANT-7. *Confirmed (partial):* Generalist bacteria (*E. coli*, *B. subtilis*) show distributed moderate rest. Specialist obligate extremophiles show different rest profiles. Persister cell literature confirms Lock-equivalent metabolic quiescence as a strategy maintained by populations under perturbation stress. Source: Trends in Microbiology (2023); Niu et al. (2024).

13. The Novel Predictions

All predictions stated before data gathered to test them. Locked here. 2026-03-18.

13.1. P-INVARIANT-1: Sleep Duration Negatively Correlates With Lineage Age Across All Phyla (Topology-Calibrated)

Cross-phyla analysis of topology-calibrated discharge duration against lineage age will show a negative correlation surviving correction for body mass, metabolic rate, and topology class. Detectable across Porifera, Cnidaria, Mollusca, Arthropoda, and Vertebrata simultaneously.

13.2. P-INVARIANT-2: Mass Extinction Survivor Lineages Show Lock or Navigator Signatures Relative to Casualties

Lock survivors: lower topology-calibrated discharge than casualties. Navigator survivors: higher dietary breadth than casualties.

13.3. P-INVARIANT-3: The Unstable Zone Boundary Is Predictable From Duration, Topology, and Dietary Breadth

Three-dimensional boundary in the space of (sleep duration, topology class, dietary breadth) separates Lock, Navigator, and Unstable Zone organisms. Topology as third axis removes the ambiguity between ectothermic Navigators (T2, low duration) and true Lock organisms that the two-dimensional v1.0 boundary could not resolve.

13.4. P-INVARIANT-4: Topology 2 Found Specifically and Exclusively in Ectothermic Navigators

T2 (distributed bilateral rest, bilateral EEG, no hemispheric asymmetry) will be found in organisms satisfying both (a) ecological generalism and (b) ectothermic metabolic architecture. Not in Lock organisms. Not in endothermic Navigators. The crocodilian is the confirmed case. Predicted additionally: monitor lizards, large ectothermic generalist fish, large iguana species.

13.5. P-INVARIANT-5 (Refined): Sleep Rebound Identifies T5, Not Basin Restructuring

Minimal-sleep organisms divide into: Class A (true Lock or basin restructuring): no rebound on pressure removal, performance maintained. Class B (T5): rebound on pressure removal, performance impaired during low-sleep period. Migratory birds are Class A during migration (basin restructuring, not debt suppression). Giraffes are Class B candidates. The performance measure during the low-sleep period is the diagnostic: maintained = restructuring; impaired = T5.

13.6. P-INVARIANT-6: Lock Organisms Are Undomesticable

No Lock organism has been or will be domesticated. The basin depth that produces perturbation-invariance renders the basin immovable by domestication fields that do not also destroy the organism.

13.7. P-INVARIANT-7: Dimensional Invariant Detectable Below Nervous System in Metabolic Rest State Duration

Generalist bacteria show moderate distributed rest. Specialist bacteria show patterns consistent with their niche-maintenance cost. Confirmed partially by persister cell literature.

13.8. P-INVARIANT-8: T6 Quiescence Duration Scales With Basin Transition Magnitude, Not Ongoing Navigational Cost

Developmental transition sleep (T6) duration will be uncorrelated with the organism's adult navigational cost profile and will instead correlate with the depth of the destination basin: a highly specialised adult organism (deep adult basin) will show longer T6 quiescence than a generalist organism undergoing the same developmental transition, because the transition into a deeper basin costs more.

Testable: Compare metamorphic quiescence duration in obligate specialists (deep adult basin) versus ecological generalists (moderate adult basin) across holometabolous insects.

13.9. P-INVARIANT-9: T7 Sleep Scales Linearly With Immune Response Depth and Will Be Maximal in Lock Organisms Relative to Baseline

T7 (immune-basin sleep) duration will scale linearly with inflammatory marker concentration, cytokine titre, or metabolic rate increase during infection.

Lock organisms, with near-zero baseline sleep, will show the largest fractional T7 sleep increase relative to baseline of any organism class when infected. A horseshoe crab infected with a pathogen will show a dramatic shift from T1 to T7, measurable as a sudden behavioural quiescence event disproportionate to any waking-state recovery need. This is testable and has not been measured.

13.10. P-INVARIANT-10: Endothermic Navigators Shifted to Ectothermy Will Shift From T3 to T2

Any endothermic organism experimentally maintained at reduced body temperature (pharmacological ectothermy or induced hibernation-like state) will shift from consolidated T3 sleep toward distributed T2-like rest as metabolic rate and ROS generation rate decrease. Torpor in hibernating mammals is a partial natural confirmation: as metabolic rate drops during torpor, sleep architecture shifts from concentrated NREM toward distributed low-activity rest. Full T2 will be expressed at full ectothermic metabolic rates.

13.11. P-INVARIANT-11: Population Persister Fraction Scales With Historical Perturbation Frequency

Bacterial populations with a history of frequent antibiotic-equivalent perturbations will maintain higher persister fractions (higher Lock sub-population) than naive populations with equivalent genetic background. This is the population-level Lock/Navigator trade-off: high perturbation frequency selects for higher Lock allocation in the population portfolio. Testable by comparing persister fractions in clinical isolates (high antibiotic exposure history) versus environmental isolates (low antibiotic exposure) of the same species.

14. The Dimensional Invariant: Complete Formal Statement

THE DIMENSIONAL INVARIANT: COMPLETE STATEMENT (v3.0)

Across all life, at all scales, across all of deep time, the framework $S + N + G \rightarrow R$ makes three inseparable predictions about rest:

Prediction 1 — Quantity: The total cost of basin occupation (Axis 1 between-basin switching cost, Axis 2 within-basin metabolic cost, Axis 3 within-basin computational intensity, plus T6 transition cost and T7 immune-basin cost when applicable) must be discharged per cycle. Duration of discharge tracks total cost within a topology class. Basin depth predicts survival horizon. The oldest surviving lineages carry the lowest cost. They discharge the least.

Prediction 2 — Topology: The architecture of the discharge process is a further geometric prediction of the same framework. Seven topologies are derived: T1 (quiescence/biochemical rhythm), T2 (distributed bilateral rest, ectothermic Navigator), T3 (consolidated bilateral deep sleep, endothermic), T4 (permanent Basin Ω , open-ocean lethal constraint), T5 (fragmented shallow, external suppression, rebound diagnostic), T6 (developmental transition cost), T7 (immune-basin occupation). Each topology is the only architecturally feasible solution for its landscape position given the full constraint set.

Prediction 3 — Substrate: The cost of basin occupation is fundamentally oxidative. Navigation generates reactive oxygen species as a metabolic byproduct at a rate proportional to the metabolic architecture and navigational intensity of the organism. Sleep discharges both the synaptic cost (slow-wave downscaling via SHY) and the oxidative cost (glymphatic pulsation clearing neurotoxic metabolites and ROS) simultaneously. The topology of sleep determines the efficiency and architecture of both discharge processes together. The feasibility of any topology is constrained by the organism's ROS generation rate relative to the clearance capacity of the topology.

The invariant holds because:

1. Basin depth is the same geometric property at every scale.
2. Navigation generates oxidative and synaptic cost at every scale.
3. Sleep — in whatever topology the landscape geometry permits — is the payment.
4. Deep-time survival is a direct function of basin depth.
5. Therefore sleep duration (topology-calibrated) is a proxy for survival trajectory.

The invariant extends below the level of the nervous system to metabolic rest states in single-celled organisms, and below the level of the individual organism to population-level basin portfolios in bacterial persister dynamics.

The oldest surviving lineages sleep the least.
How they sleep is the geometry of where they are.
What they are paying is an oxidative debt.
This is not a coincidence.
This is the geometry.

15. The Unified Derivation Chain

Step	Claim	Record
1. Cancer	FOXA1/EZH2 ratio identifies whether a cancer cell is epigenetically arrested in the cancer attractor.	10.5281/zenodo.18883922
2. Domestication	The domestication syndrome is field-specified developmental arrest of NCC migration. The pig is the clean case.	10.5281/zenodo.19094935
3. Canine	Above a genomic fixation threshold, the landscape is remodelled. Basin A becomes inaccessible.	Repository 2026-03-18
4. Feline	The symmetric dual-basin landscape. Three-class domestication taxonomy. Basin Ω as meta-attractor.	Repository 2026-03-18
5. Sleep cost	Sleep duration tracks total basin- occupation cost. Three axes. Fundamentally oxidative substrate.	Repository 2026-03-18
6. Sleep sonar	Sleep spectrum maps onto the three-axis framework across all animals.	Repository 2026-03-18
7. v1.0	The dimensional invariant. Sleep predicts survival across deep time. Oldest surviving lineages sleep least.	10.5281/zenodo.19094935
8. v2.0	Sleep topology is a geometric prediction of landscape position. Five topologies. Full three-channel sonar.	10.5281/zenodo.19101495
9. v3.0	Oxidative cost substrate identified. Two new topologies (T6, T7). Population-level basin portfolio. Local sleep as circuit-level application. Eleven predictions.	This document

THE UNIFIED STATEMENT

The attractor geometry framework is a single geometric principle operating identically at every scale at which life organises itself.

From a cancer cell deciding whether to differentiate or proliferate,
to a bacterial persister cell holding the Lock position in a population portfolio while its neighbours navigate,
to a *C. elegans* larva that cannot complete its developmental transition without mandatory quiescence paying the transition cost,
to a crocodile distributed across 16 hours of bilateral rest because its ectothermic ROS rate makes concentrated sleep unnecessary and ecologically dangerous,
to a dolphin that has dissolved the barrier between waking and sleep because the ocean makes full bilateral sleep lethal,
to an octopus that independently evolved two-phase sleep from a completely different neural architecture because it is a Navigator paying the same navigational cost,
to a horseshoe crab that has been in the same basin for 450 million years, generating almost no oxidative

debt, and sleeping not at all:

the geometry is the same.
The basin is the same concept.
The cost of navigation is the same cost.
The oxidative debt is the same debt.
The sleep is the same payment.
The topology is the shape of the landscape.
The survival is the depth.

Scale is the only variable.
The geometry is invariant.

Linked Records

Record	DOI / Location
Theory paper	doi:10.5281/zenodo.19094935
v1.0 invariant	doi:10.5281/zenodo.19094935
v2.0 invariant	doi:10.5281/zenodo.19101495
P7 pre-registration	doi:10.5281/zenodo.19096128
P9 pre-registration	[DOI pending]
FOXA1/EZH2 cancer	doi:10.5281/zenodo.18883922
Plant inversion	doi:10.5281/zenodo.19040399
LECA resurrection	doi:10.5281/zenodo.18986790
Canine / feline / sleep artifacts	github.com/Eric-Robert-Lawson/ attractor-oncology (2026-03-18)

Document ID: DIMENSIONAL_INVARIANT_SLEEP_SURVIVAL_REASONING_ARTIFACT_v3.0

Date: 2026-03-18

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Status: REASONING ARTIFACT — DERIVATION RECORD. Self-contained. Does not rely on v1.0 or v2.0.

Primary theoretical contributions: (1) The dimensional invariant with three inseparable predictions: quantity, topology, and oxidative substrate. (2) Two-strategy taxonomy (Lock, Navigator) with unstable zone. (3) Seven sleep topologies as geometric predictions. (4) Oxidative cost substrate: navigation generates ROS; sleep clears both synaptic and oxidative debt simultaneously. (5) T6 transition-cost sleep and T7 immune-basin sleep as new topology classes. (6) Population-level basin portfolio: persister cells as Lock/Navigator allocation within a clonal population. (7) Local sleep as circuit-level application of the dimensional invariant. (8) Full three-channel sonar plus supplementary ROS rate channel. (9) Eleven predictions P-INVARIANT-1 through P-INVARIANT-11.

Prior versions: v1.0: [doi:10.5281/zenodo.19094935](https://doi.org/10.5281/zenodo.19094935); v2.0: [doi:10.5281/zenodo.19101495](https://doi.org/10.5281/zenodo.19101495)

Theory paper: [doi:10.5281/zenodo.19094935](https://doi.org/10.5281/zenodo.19094935)

Repository: [github.com/Eric-Robert-Lawson/ attractor-oncology](https://github.com/Eric-Robert-Lawson/attractor-oncology)

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Version: 3.0 — LOCKED AT PRE-REGISTRATION.